COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212- Automating System Administration	**SECTION:**	CPE31S2
DATE PERFORMED:	8/13/2026	**DATE SUBMITTED:**	8/13/2026
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ACTIVITY NO. ___ **Prelim Examination: Installing Ansible, Building an Inventory, Using Metagroups, Variables, and Creating a Playbook**

1. PROCEDURE OUTPUT

Intended Learning Outcomes Addressed:
At the end of the activity the students will be able to:
- **IInstall and verify Ansible on a control node.**
- **Create and verify an inventory file.**
- **Organize hosts using groups and metagroups.**
- **Create host and group variables.**
- **Execute Ansible ad hoc commands.**
- **Create and run a simple Ansible playbook.**

Objectives:
This laboratory guides the students to Install Community Enterprise OS and set up SSH key-based authentication
between local and remote machines.

1. Set up
The laboratory environment consists of:
- **1 Ubuntu Control Node**
- **2 Ubuntu Managed Nodes**
- **1 CentOS Managed Node**

Replace the sample IP addresses below with the actual IP addresses of your virtual machines.

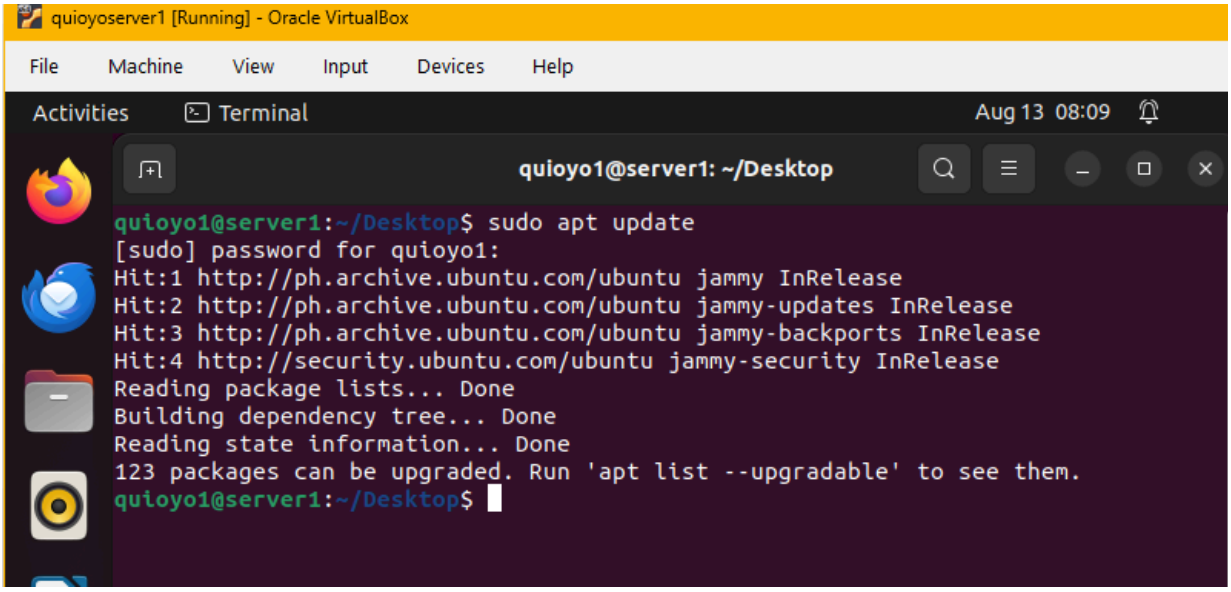
Hostname	Operating System	Ip Address
Ubuntu1	Ubuntu	192.168.56.109
Ubuntu2	Ubuntu	192.168.56.110
CentOS1	Centos	

2. MATERIALS AND EQUIPMENT
- **VirtualBox**
- **Personal Computer**

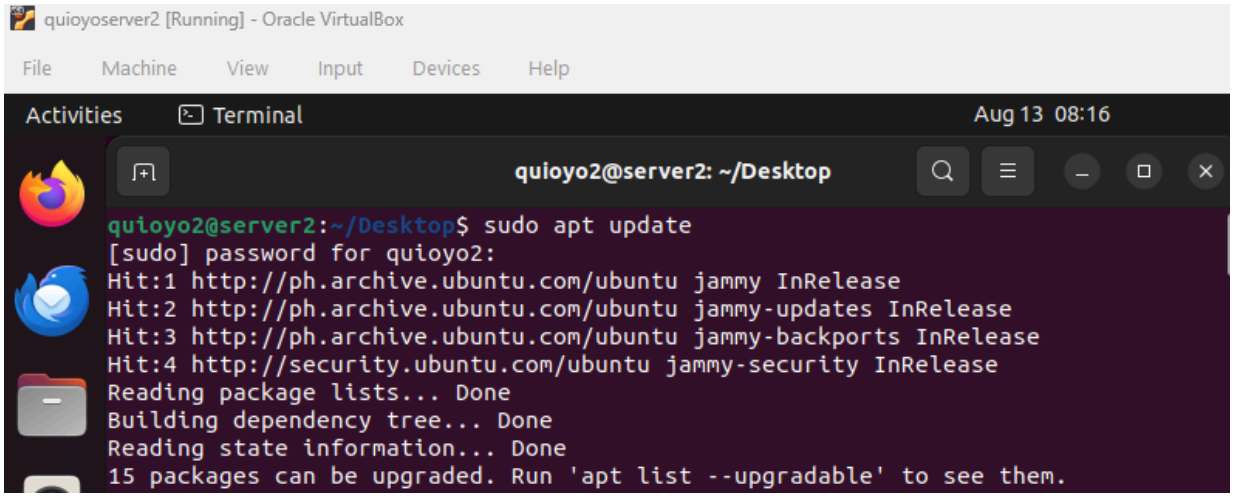
3. PROCEDURE
Part A: Installing and Verifying Ansible

1. Update the package repository

Server1:



Server2:



2. Install pip if it is not yet installed.

Server1:

```
quioyo1@server1:~/Desktop$ sudo apt install python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-11 gcc gcc-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils
  libc-dev-bin libc-devtools libc6 libc6-dbg libc6-dev libcc1-0 libcrypt-dev
  libctf-nobfd0 libctf0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc
  libjs-underscore liblsan0 libnsl-dev libpython3-dev libpython3.10-dev
  libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0 libubsan1 linux-libc-dev
  lto-disabled-list make manpages-dev python3-dev python3-distutils
  python3-setuptools python3-wheel python3.10-dev rpcsvc-proto zlib1g-dev
Suggested packages:
  binutils-doc debian-keyring g++-multilib g++-11-multilib gcc-11-doc
  gcc-multilib autoconf automake libtool flex bison gcc-doc gcc-11-multilib
  gcc-11-locales apache2 | lighttpd | httpd glibc-doc git bzr libstdc++-11-doc
  make-doc python-setuptools-doc
Recommended packages:
  libnss-nis libnss-nisplus
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-11 gcc gcc-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0
  libctf0 libdpkg-perl libexpat1-dev libfakeroot libfile-fcntllock-perl
  libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0
  libnsl-dev libpython3-dev libpython3.10-dev libquadmath0 libstdc++-11-dev
  libtirpc-dev libtsan0 libubsan1 linux-libc-dev lto-disabled-list make
  manpages-dev python3-dev python3-distutils python3-pip python3-setuptools
  python3-wheel python3.10-dev rpcsvc-proto zlib1g-dev
The following packages will be upgraded:
```

Server2:

```
quioyo2@server2:~/Desktop$ sudo apt install python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
  linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic
  linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-11 gcc gcc-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0
  libctf0 libdpkg-perl libexpat1-dev libfakeroot libfile-fcntllock-perl
  libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0
  libnsl-dev libpython3-dev libpython3.10-dev libquadmath0 libstdc++-11-dev
  libtirpc-dev libtsan0 libubsan1 linux-libc-dev lto-disabled-list make
  manpages-dev python3-dev python3-distutils python3-setuptools python3-wheel
  python3.10-dev rpcsvc-proto zlib1g-dev
Suggested packages:
  binutils-doc debian-keyring g++-multilib g++-11-multilib gcc-11-doc
  gcc-multilib autoconf automake libtool flex bison gcc-doc gcc-11-multilib
  gcc-11-locales apache2 | lighttpd | httpd glibc-doc git bzr libstdc++-11-doc
  make-doc python-setuptools-doc
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-11 gcc gcc-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0
  libctf0 libdpkg-perl libexpat1-dev libfakeroot libfile-fcntllock-perl
  libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0
  libnsl-dev libpython3-dev libpython3.10-dev libquadmath0 libstdc++-11-dev
  libtirpc-dev libtsan0 libubsan1 linux-libc-dev lto-disabled-list make
  manpages-dev python3-dev python3-distutils python3-pip python3-setuptools
  python3-wheel python3.10-dev rpcsvc-proto zlib1g-dev
0 upgraded, 53 newly installed, 0 to remove and 15 not upgraded.
```

3. Install Ansible.
Server1:

```
quioyo1@server1:~/Desktop$ python3 -m pip install --user ansible
Collecting ansible
  Downloading ansible-10.7.0-py3-none-any.whl (51.6 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.6/51.6 MB 2.3 MB/s eta 0:00:00
Collecting ansible-core~=2.17.7
  Downloading ansible_core-2.17.14-py3-none-any.whl (2.2 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.2/2.2 MB 2.7 MB/s eta 0:00:00
Requirement already satisfied: PyYAML>=5.1 in /usr/lib/python3/dist-packages (fr
om ansible-core~=2.17.7->ansible) (5.4.1)
Requirement already satisfied: cryptography in /usr/lib/python3/dist-packages (f
rom ansible-core~=2.17.7->ansible) (3.4.8)
Collecting packaging
  Downloading packaging-26.3-py3-none-any.whl (129 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 130.0/130.0 KB 3.2 MB/s eta 0:00:00
Collecting jinja2>=3.0.0
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 KB 7.0 MB/s eta 0:00:00
Collecting resolvelib<1.1.0,>=0.5.3
  Downloading resolvelib-1.0.1-py2.py3-none-any.whl (17 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/lib/python3/dist-packages
 (from jinja2>=3.0.0->ansible-core~=2.17.7->ansible) (2.0.1)
Installing collected packages: resolvelib, packaging, jinja2, ansible-core, ansi
ble
  WARNING: The scripts ansible, ansible-config, ansible-connection, ansible-cons
ole, ansible-doc, ansible-galaxy, ansible-inventory, ansible-playbook, ansible-p
ull and ansible-vault are installed in '/home/quioyo1/.local/bin' which is not o
n PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warn
ing, use --no-warn-script-location.
  WARNING: The script ansible-community is installed in '/home/quioyo1/.local/bi
n' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warn
ing, use --no-warn-script-location.
Successfully installed ansible-10.7.0 ansible-core-2.17.14 jinja2-3.1.6 packagin
```

Server2:

```
quioyo2@server2:~/Desktop$ sudo apt install ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
  linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic
  linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ieee-data python-babel-localedata python3-argcomplete python3-babel
  python3-dnspython python3-jinja2 python3-jmespath python3-kerberos
  python3-libcloud python3-netaddr python3-ntlm-auth python3-packaging
  python3-pycryptodome python3-requests-kerberos python3-requests-ntlm
  python3-requests-toolbelt python3-selinux python3-simplejson python3-winrm
  python3-xmltodict
Suggested packages:
  cowsay sshpass python3-sniffio python3-trio python-jinja2-doc ipython3
  python-netaddr-docs
The following NEW packages will be installed:
  ansible ieee-data python-babel-localedata python3-argcomplete python3-babel
  python3-dnspython python3-jinja2 python3-jmespath python3-kerberos
  python3-libcloud python3-netaddr python3-ntlm-auth python3-packaging
  python3-pycryptodome python3-requests-kerberos python3-requests-ntlm
  python3-requests-toolbelt python3-selinux python3-simplejson python3-winrm
  python3-xmltodict
0 upgraded, 21 newly installed, 0 to remove and 15 not upgraded.
Need to get 28.0 MB of archives.
After this operation, 271 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 python-babel-localeda
ta all 2.8.0+dfsg.1-7 [4,982 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 python3-babel all 2.8
.0+dfsg.1-7 [85.1 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-jinja
2 all 3.0.3-1ubuntu0.4 [108 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 python3-packaging all
```

4. Verify that Ansible is installed correctly.
Server1:

```
quioyo1@server1:~/Desktop$ ansible --version
ansible [core 2.17.14]
  config file = None
  configured module search path = ['/home/quioyo1/.ansible/plugins/modules', '/u
sr/share/ansible/plugins/modules']
  ansible python module location = /home/quioyo1/.local/lib/python3.10/site-pack
ages/ansible
  ansible collection location = /home/quioyo1/.ansible/collections:/usr/share/an
sible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/usr/bin/
python3)
  jinja version = 3.1.6
  libyaml = True
```

Server2:

```
quioyo2@server2:~/Desktop$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/quioyo2/.ansible/plugins/modules', '/u
sr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
```

5. Verify SSH connectivity to each managed node.

Server1:

```
quioyo1@server1:~/Desktop$ ssh quioyo1@192.168.56.109
quioyo1@192.168.56.109's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

128 updates can be applied immediately.
123 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

3 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Thu Jul 30 10:17:11 2026 from 192.168.56.108
```

Server2:

```
quioyo2@server2:~/Desktop$ ssh quioyo2@192.168.56.110
quioyo2@192.168.56.110's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

15 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

3 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 30 10:17:54 2026 from 192.168.56.108
```

Part B: Building an Inventory File
inventories provide Ansible with information about managed nodes and their network locations.

1. Create a working directory.
Server1:

```
quioyo1@server1:~$ mkdir ~/ansible-lab
quioyo1@server1:~$ cd ~/ansible-lab
```

quioyo1@server1: ~/ansible-lab

Server2:

```
quioyo2@server2:~$ mkdir ~/ansible-lab
quioyo2@server2:~$ cd ~/ansible-lab
```

quioyo2@server2: ~/ansible-lab

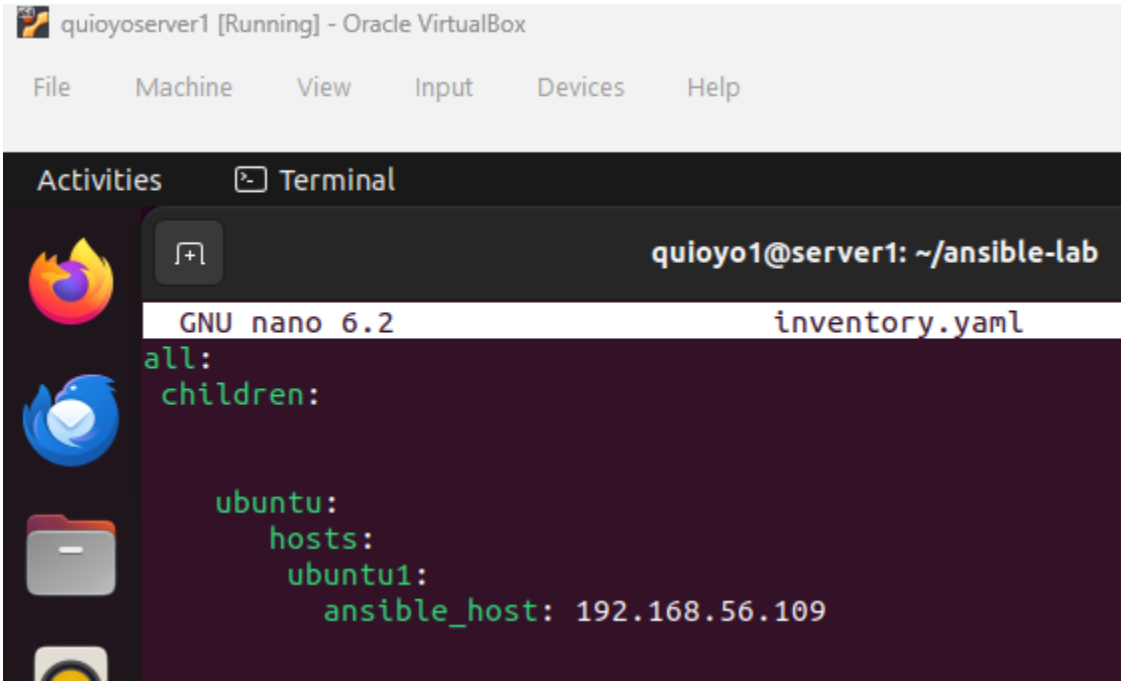
2. Create an inventory file.
Server1:

```
quioyo1@server1:~/ansible-lab$ nano inventory.yaml
```

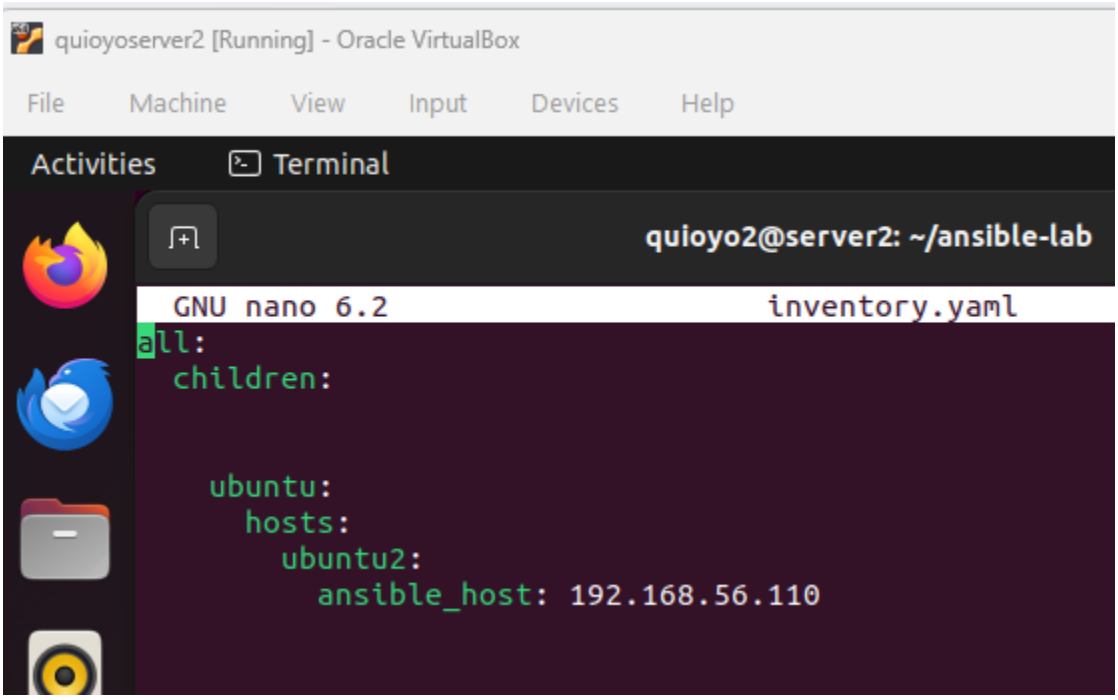
Server2:

```
quioyo2@server2:~/ansible-lab$ nano inventory.yaml
```

3. Add the following content.
Server1:



Server2:



4. Verify the inventory structure
Server1:

```
quioyo1@server1:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "ubuntu1": {
                "ansible_host": "192.168.56.109"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1"
        ]
    }
}
```

```
quioyo1@server1:~/ansible-lab$ ansible all --list-hosts -i inventory.yaml
  hosts (2):
    ubuntu1
    ubuntu2
```

Server2:

```
quioyo2@server2:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "ubuntu2": {
                "ansible_host": "192.168.56.110"
            }
        }
    },
    "all": {
        "children": [
            "ubuntu",
            "ungrouped"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu2"
        ]
    }
```

```
quioyo2@server2:~/ansible-lab$ ansible all --list-host -i inventory.yaml
  hosts (2):
    ubuntu2
    ubuntu1
```

5. Ping all managed nodes.

Server1:

```
quioyo1@server1:~/ansible-lab$ ansible all -m ping -i inventory.yaml
[WARNING]: Platform linux on host ubuntu1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host ubuntu2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
```

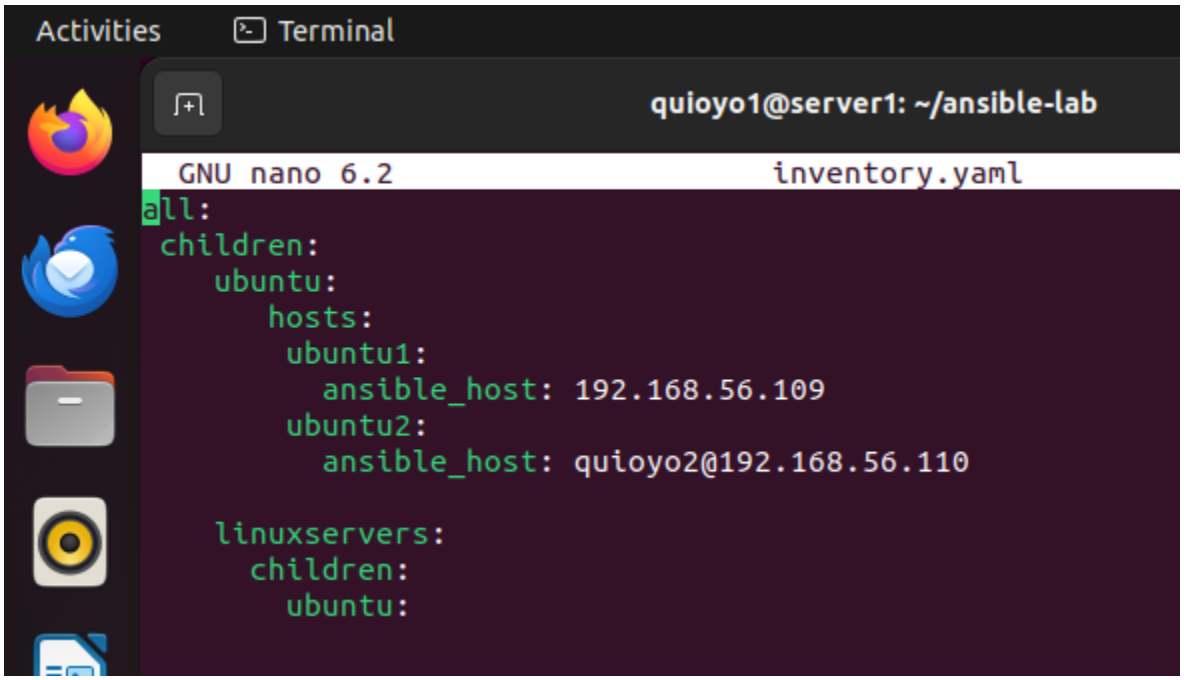
Server2:

```
quioyo2@server2:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

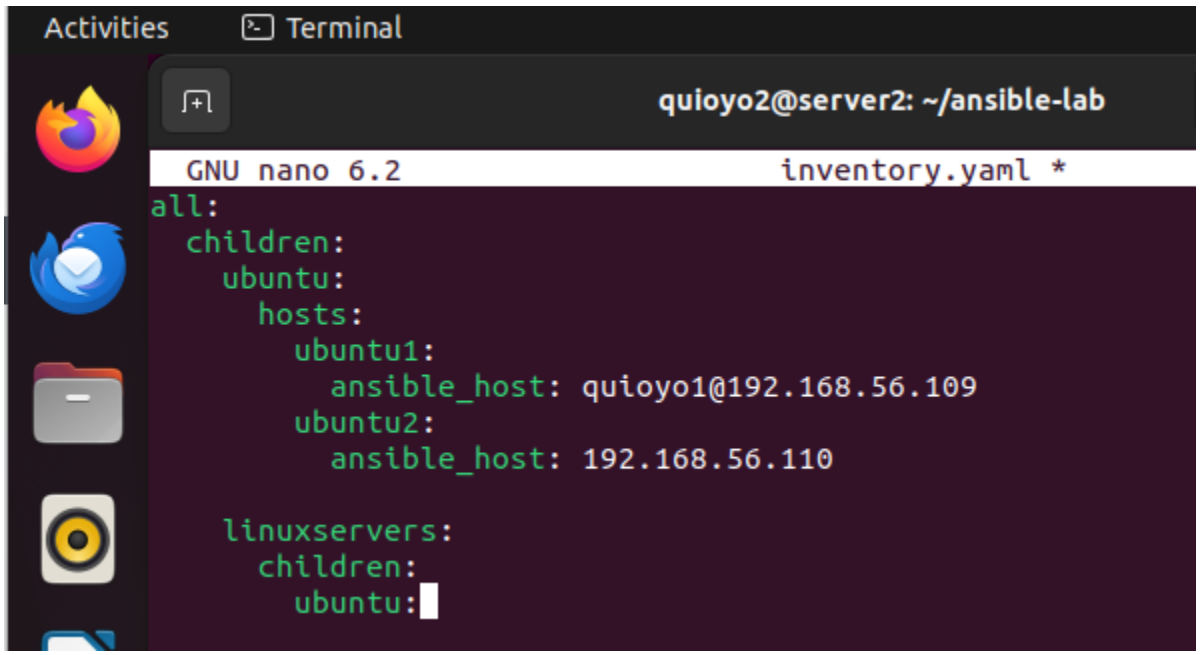
Part C: Creating a Metagroup

Metagroups allow multiple groups to be combined into a single logical group.

1. Modify the inventory file.
Server1:



Server2:



2. Test the metagroup.

Server1:

```
quioyo1@server1:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
[WARNING]: Platform linux on host ubuntu1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host ubuntu2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
```

Server2:

```
quioyo2@server2:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Trash

3. Display all hosts belonging to the metagroup.

Server1:

```
quioyo1@server1:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.y
ml
  hosts (2):
    ubuntu1
    ubuntu2
```

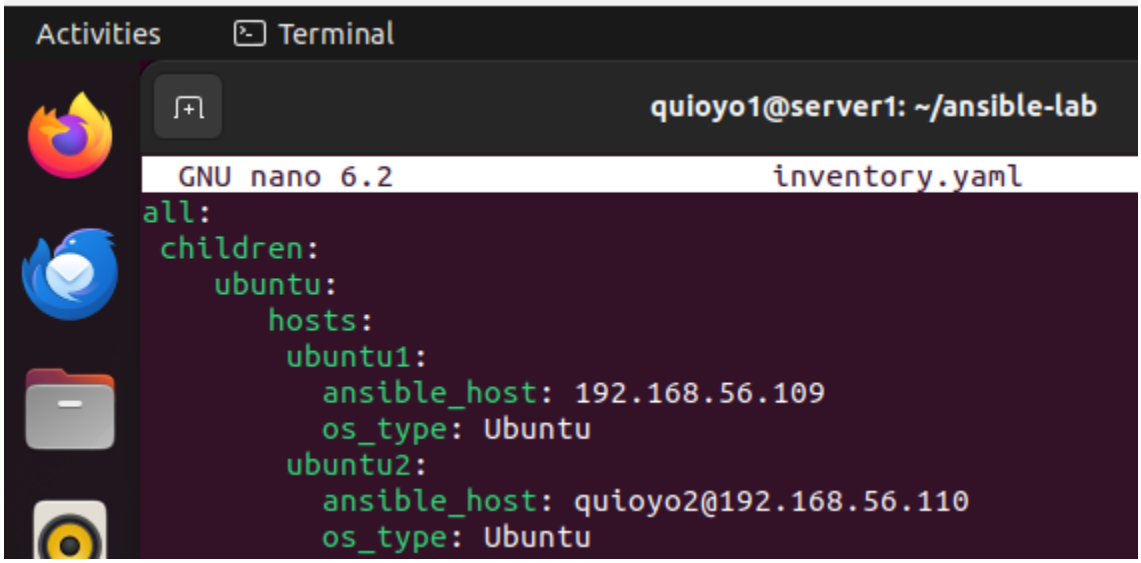
Server2:

```
quioyo2@server2:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.ya
ml
  hosts (2):
    ubuntu1
    ubuntu2
```

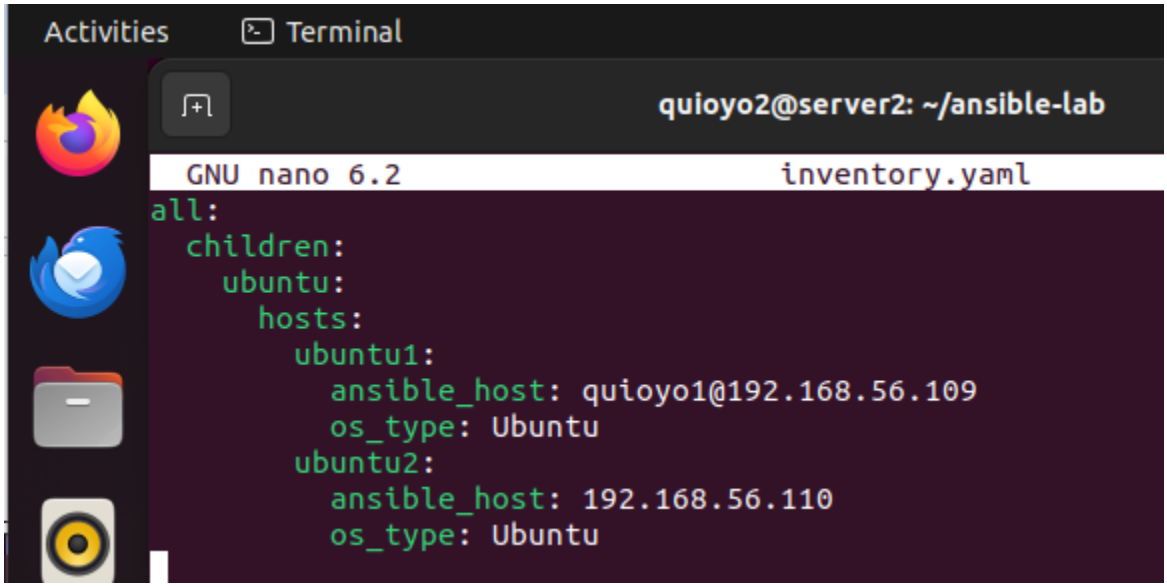
Part D: Creating Variables

Variables allow information such as operating systems and usernames to be assigned to hosts and groups.

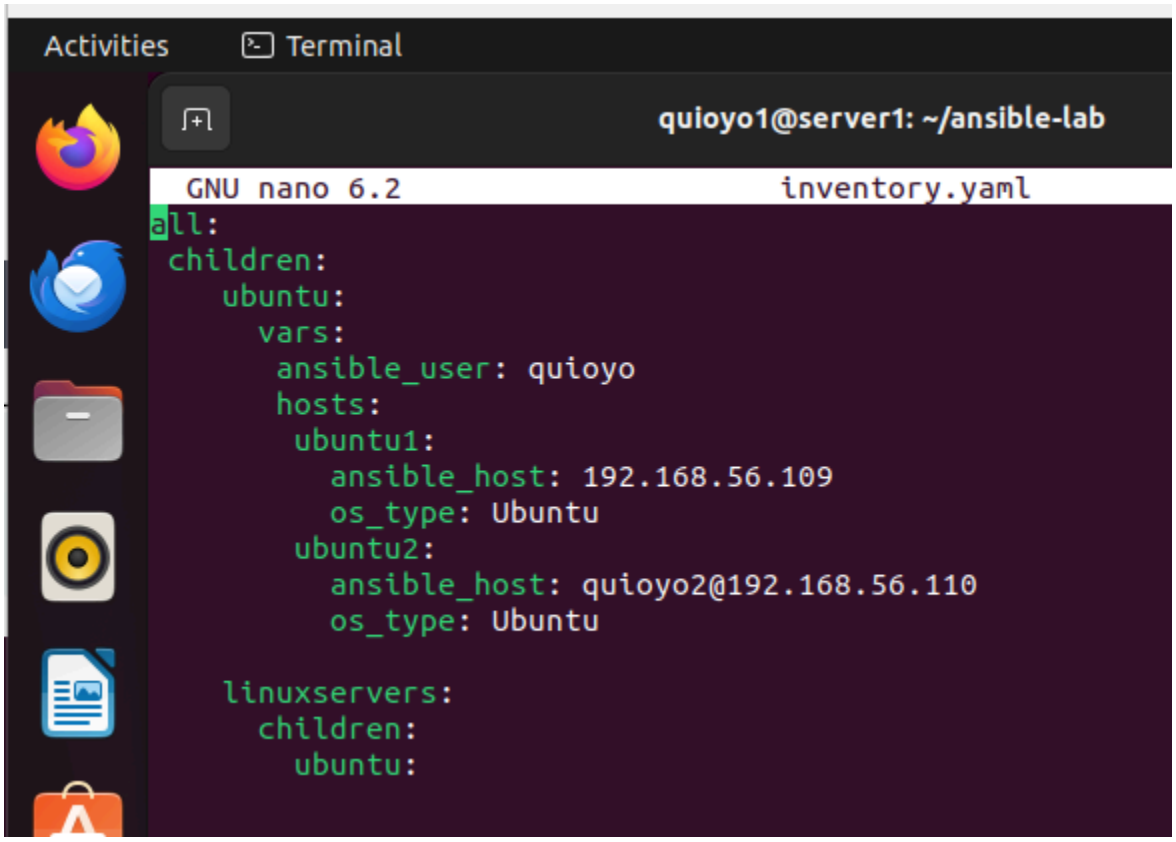
1. Add host variables.
Server1:



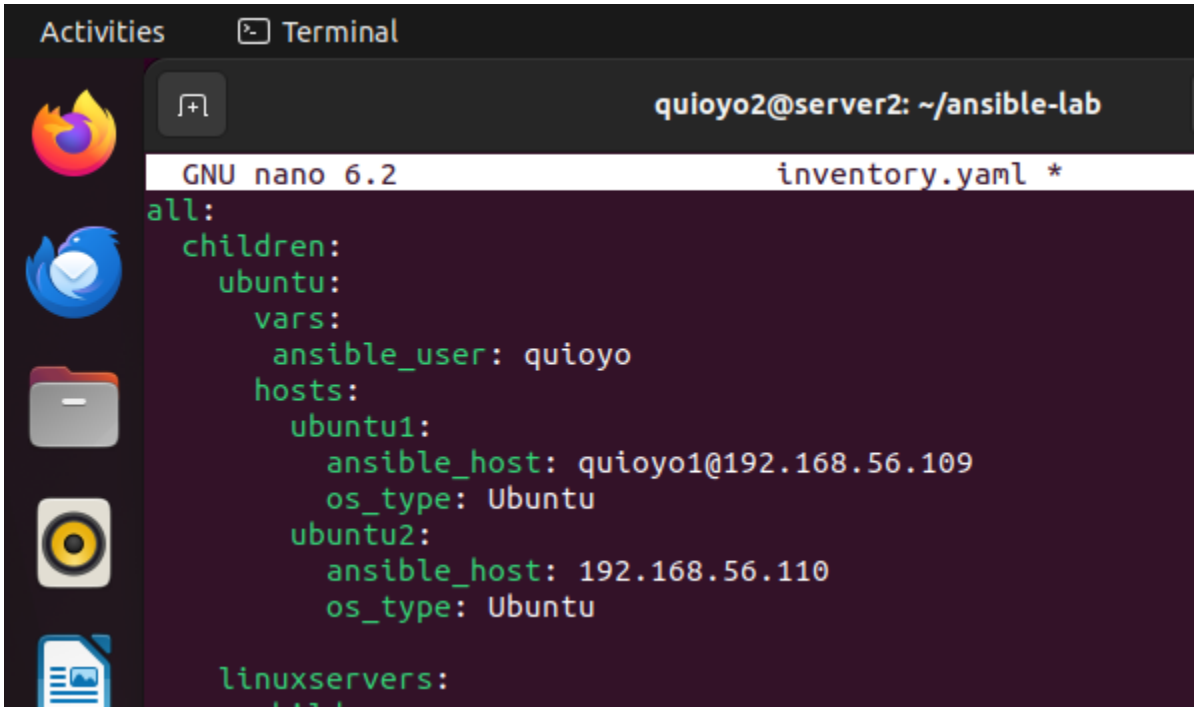
Server2:



2. Add group variables.
Server1:



Server2:



3. Create a playbook to display variables.
Server1:

Server2:

Conclusion:
In conclusion, this prelim skills exam is all about the ansible where we set the ip address, copy the ssh id and istall ansible. It creates tasks and by using yaml and using playsboooks.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

COMPUTER ENGINEERING

QUEZON CITY

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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